# Converting to & from Standard Form

**Exam code:** 0607

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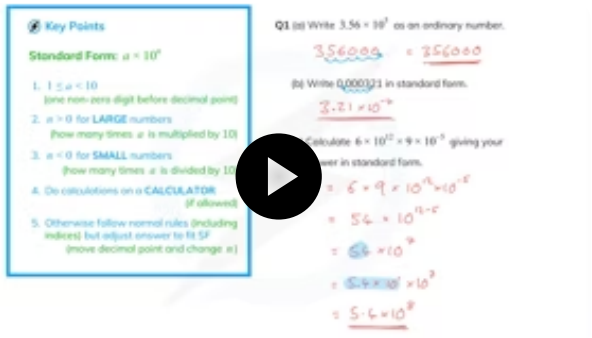


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## Converting to and from standard form

### What is standard form and why is it used?

- **Standard form** is a way of writing **numbers** using **powers of 10**
- It is helpful to see the size of **very large** and **very small** numbers
- This allows us to:
  - Write them more concisely
  - Compare them more easily
  - Perform calculations with them more easily

### How do I write a number in standard form?

- Numbers written in standard form are always written as:

$$a \times 10^n$$

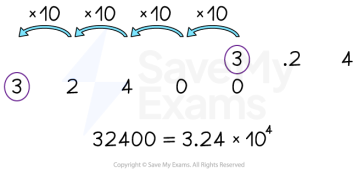
- Where:
  - $1 \le a < 10$ ($a$ is **bigger than or equal to 1 and less than 10**)
  - $n$ is an **integer**

**Stuck on something?**
Ask questions and get help with exactly what you're revising.
Got it

less than 10
    - $n > 0$ ($n$ is **positive**) for numbers bigger than or equal to 10

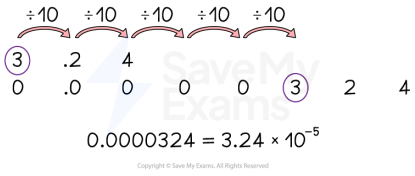
## How do I write a large number in standard form?

- To write a **large number** such as 32 400 in standard form
  - Identify the value of $a$
    - 3.24
  - Find **how many times** you must **multiply** 3.24 **by 10**, to make 32 400
    - Count how many places you need to move the digits
    - We need to multiply by 10 four times
  - 32 400 = 3.24 × 10 × 10 × 10 × 10 = **3.24 × 10$^4$**




## How do I write a small number in standard form?

- To write a **small number** such as 0.0000324 in standard form
  - Identify the value of $a$
    - 3.24
  - Find **how many times** you must **divide** 3.24 **by 10**, to make 0.0000324
    - Count how many places you need to move the digits
    - We need to divide by 10 five times
    - We are dividing rather than multiplying so the **power will be negative**
  - 0.0000324 = 3.24 ÷ 10 ÷ 10 ÷ 10 ÷ 10 ÷ 10 = **3.24 × 10$^{-5}$**




### Examiner Tips and Tricks

## Worked Example

(a) Without a calculator, write 0.007052 in standard form.

**Answer:**

Standard form will be written as $a \times 10^n$ where $a$ is between 1 and 10
Find the value for $a$

$$a = 7.052$$

The original number is smaller than 1 so $n$ will be negative
Count how many times you need to divide $a$ by 10 to get the original number

$$0.007052 = 7.052 \div 10 \div 10 \div 10 \text{ (3 times)}$$

Therefore $n = -3$.

$$\mathbf{0.007052 = 7.052 \times 10^{-3}}$$

(b) Without a calculator, write 324 500 000 in standard form.

**Answer:**

Standard form will be written as $a \times 10^n$ where $a$ is between 1 and 10
Find the value for $a$

$$a = 3.245$$

The original number is larger than 1 so $n$ will be positive
Count how many times you need to multiply $a$ by 10 to get the original number

$$324\,500\,000 = 3.245 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10 \text{ (8 times)}$$

Therefore $n = 8$

$$\mathbf{324\,500\,000 = 3.245 \times 10^8}$$

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